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it must accelerate or decelerate respectively. If this suddenly happens it makes the rolling object to slip or slide.

*Sliding*

Sliding is the case when $v_{CM} > R\omega$ (or $v_{TRANS} > v_{ROT}$). The translation is more than the rotation. This kind of motion happens when sudden break is applied in a moving vehicles, or when the vehicle enters into a slippery road. In this case, the point of contact has more of $v_{TRANS}$ than $v_{ROT}$. Hence, it has a resultant velocity v in the forward direction as shown in Figure 5.35. The kinetic frictional force ($f_k$) opposes the relative motion. Hence, it acts in the opposite direction of the relative velocity. This frictional force reduces the translational velocity and increases the rotational velocity till they become equal and the object sets on pure rolling. Sliding is also referred as forward slipping.

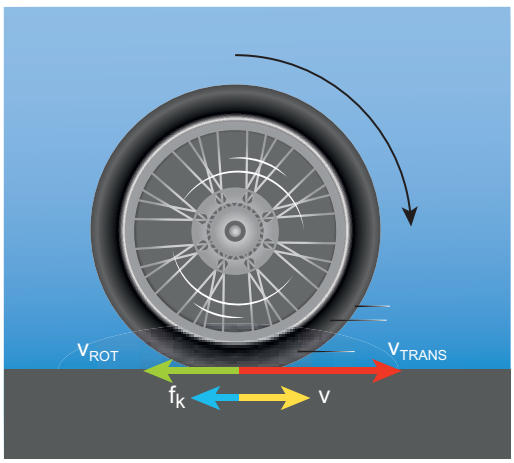



**Figure 5.35** Sliding

*Slipping*

Slipping is the case when $v_{CM} < R\omega$ (or $v_{TRANS} < v_{ROT}$). The rotation is more than the translation. This kind of motion happens when we suddenly start the vehicle from rest or the vehicle is stuck in mud. In this case, the point of contact has more of $v_{ROT}$ than $v_{TRANS}$. It has a resultant velocity v in the backward direction as shown in Figure 5.36. The kinetic frictional force ($f_k$) opposes the relative motion. Hence it acts in the opposite direction of the relative velocity. This frictional force reduces the rotational velocity and increases the translational velocity till they become equal and the object sets pure rolling. Slipping is sometimes empahasised as backward slipping.

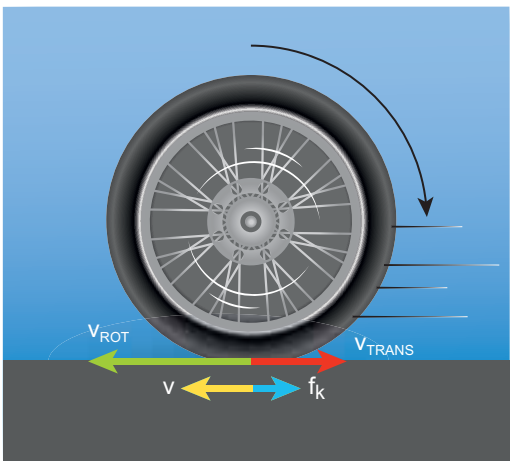



**Figure 5.36** Slipping

3ATRR1

**EXAMPLE 5.21**

A rolling wheel has velocity of its center of mass as 5 $m\,s^{-1}$. If its radius is 1.5 m and angular velocity is 3 rad $s^{-1}$, then check whether it is in pure rolling or not.